AI MCQ Generator - Quiz Report

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Score: 2/10

Percentage: 20.00%

Question 1: According to the study material, what is the precise distinction between a Smart Sensor and an Intelligent Sensor?

A. A smart sensor has an analog output, whereas an intelligent sensor has a quasidigital output.
B. An intelligent sensor is a smart sensor integrated with a Digital Processor (DPU).
C. A smart sensor requires external power, while an intelligent sensor is completely self-generating.
D. An intelligent sensor is fabricated with a wire-wound core, whereas a smart sensor uses a thin film.
Your Answer: A
Correct Answer: B
Incorrect
Explanation: The study material defines a Smart Sensor as a module containing sensor elements integrated with necessary electronics in a single IC chip, and defines an Intelligent Sensor as a 'Smart sensor + Digital Processor (DPU)'.

Question 2: Which of the following is explicitly classified as an Active Transducer in the provided study material?

A. Linear Variable Differential Transformer (LVDT)
B. Resistive potentiometer (POT)
C. Si-type Diaphragm Pressure sensor (with a strain gauge and a thin film resistor network)
D. Capacitive displacement transducer
Your Answer: A
Correct Answer: C
Incorrect
Explanation: The study material states that active transducers are self-generating and draw power from the applied input. It explicitly lists 'Si-type Diaphragm Pressure sensor (with the strain gauge and a thin film resistor network)' as an example of an Active Transducer, while resistive, inductive, and capacitive transducers are classified as passive.

Question 3: Under what condition does a thermistor achieve high precision, and what is its typical temperature span around the target temperature?

A. Within a limited temperature range of about 50ºC, depending on the base resistance.
B. Across a broad temperature range of -200 °C to +850 °C, depending on the excitation current.
C. Within exactly 100ºC, depending on the thickness of the impermeable epoxy encapsulation.
D. Within a range of 10ºC, depending on the thermal time delay of the self-heating switch.
Your Answer: A
Correct Answer: A

Correct
Explanation: The study material states: 'Typically, a thermistor achieves high precision within a limited temperature range of about 50ºC around the target temperature. This range is dependent on the base resistance.'

Question 4: What is the typical operating temperature range of Resistance Temperature Detectors (RTDs), and what type of temperature coefficient do their pure metals exhibit?

A. -50 °C to +350 °C with a negative temperature coefficient of resistance
B. -200 °C to +850 °C with a positive temperature coefficient of resistance
C. -100 °C to +500 °C with a negative temperature coefficient of resistance
D. 0 °C to +1000 °C with a positive temperature coefficient of resistance
Your Answer: A
Correct Answer: B
Incorrect
Explanation: The study material states that RTDs can operate over a large temperature range from -200 °C to +850 °C and that pure metals typically have a positive temperature coefficient of resistance.

Question 5: How is a thermistor utilized to act as a slow switch in time delay applications according to the study material?

A. Through an external mechanical force that shifts the sliding contact.
B. By utilizing self-heating from a large current which 'opens' the thermistor.
C. By changing the dielectric constant using liquid level variations.
D. By applying an AC excitation voltage to neutralize the base resistance.
Your Answer: A
Correct Answer: B
Incorrect
Explanation: The study material states that one of the thermistor's applications is 'Time delay (self heating from large current ‘opens’ the thermistor so it can be used as a slow switch).'

Question 6: In a Rotary Variable Differential Transformer (RVDT), what happens when the core undergoes counter-clockwise rotation from the primary null position?

A. The output voltage becomes zero because s1 and s2 voltages are in phase.
B. It produces an increasing voltage of the secondary winding of the opposite phase.
C. It reduces the self-inductance of the primary winding to zero.
D. It results in an instantaneous conversion of physical displacement to a quasidigital pulse.
Your Answer: A
Correct Answer: B
Incorrect
Explanation: According to the study material, for an RVDT, 'Clockwise rotation produces an increasing voltage of secondary winding of one phase while counter clockwise rotation produces an increasing voltage of opposite phase.'

Question 7: For a capacitive transducer using a change in the area of plates to measure linear displacement, what is its suitable displacement measurement range and achievable accuracy?

A. Displacement ranging from 1mm to 10mm, with accuracy as high as 0.005%.
B. Displacement ranging from 10mm to 100mm, with accuracy as high as 0.05%.
C. Displacement ranging from 1mm to several cms, with accuracy of exactly 1.0%.
D. Displacement ranging from 100mm to 1000mm, with accuracy as high as 0.001%.
Your Answer: A
Correct Answer: A
Correct
Explanation: The study material states that this type of capacitor (change in plate area for linear displacement) is suitable for measurement of linear displacements ranging from 1mm - 10mm and that its accuracy is as high as 0.005%.

Question 8: What defines a passive transducer according to the classification rules in the study material?

A. It converts electrical signals directly to physical quantities without moving parts.
B. It generates its own electrical signal directly from the measured physical quantity.
C. It derives the power required for transduction from an auxiliary, external power source.
D. It must contain a digital processor (DPU) and act as a smart single IC chip.
Your Answer: A
Correct Answer: C
Incorrect
Explanation: The study material defines a Passive Transducer as a 'Device which derive power required for transduction from auxiliary power source' and notes they are 'externally powered'.

Question 9: Why are resistance strain gauges also known as piezoresistive gauges?

A. Because their capacitance changes linearly with overlapping plate area.
B. Because there is a change in the value of resistivity of the conductor when it is strained.
C. Because they generate self-excited voltage pulses without external power.
D. Because they work on the principle of variable mutual inductance.
Your Answer: A
Correct Answer: B
Incorrect
Explanation: The study material states: 'Also there is a change in the value of resistivity of the conductor when it is strained and this property is called piezoresistive effect. Therefore, resistance strain gauges are also known as piezoresistive gauges.'

Question 10: In the provided RTD working example, how is the proportional voltage signal generated to allow the calculation of temperature?

A. By applying an AC voltage that varies the mutual inductance of a ceramic core.
B. By passing an excitation current (I1) through the temperature-dependent resistance of the sensor.
C. By moving a sliding wiper contact across a pure platinum-plated wire.
D. By utilizing the self-generating property of the wire-wound glass core.
Your Answer: A

Correct Answer: B
Incorrect
Explanation: According to the RTDs working example in the study material: 'An excitation current, I1, is passed through the temperature-dependent resistance of the sensor. This produces a voltage signal that is proportional to both the excitation current and the RTD resistance.'